ECO 362: Financial Investments
Diversification

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A tale of two friends

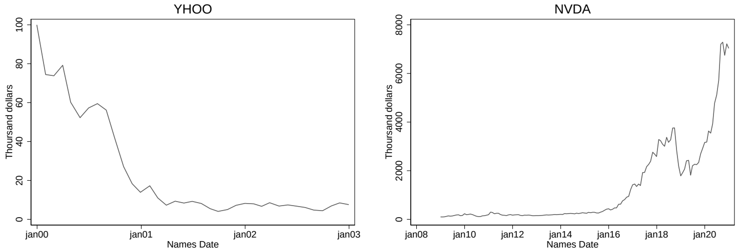
1. Janet invested \$100k in a single stock, Yahoo (YHOO), in December 1999.

Number of unique visitors in January 2000

Rank	Site	millions
1	yahoo.com	36.820
2	aol.com	36.691
3	geocities.com	28.492
4	msn.com	26.746
5	lycos.com	21.251

2. Ben invested \$100k in a single stock, Nvidia (NVDA), in December 2008.

How did they do?



Common investing mistake

- Outcome bias: Evaluate decision based on outcome, which ultimately depends on luck, rather than information known at the time that the decision was made.
- Janet thinks that stocks are risky and vouches to never touch stocks again.
- Ben thinks that he is an investment genius.
- They both fail this lecture in ECO 362. Let's learn why!

Questions

- How do you construct an optimal portfolio of risky assets?
  - Over 3,000 publicly listed US stocks.
  - MSCI World Index is a portfolio of 23 developed stock markets.
- Should you simply hold the market portfolio (weighted by market cap)?
- Can you do better?

Benefits of diversification with two risky assets

- Asset 1: R_1 has mean μ and variance σ^2.
- Asset 2: R_2 has mean μ and variance σ^2.
- Correlation between the assets is $\rho = \mathrm{Corr}(R_1, R_2)$.
- Construct an equal-weighted portfolio of the two assets as

$$R_p = \frac{1}{2}(R_1 + R_2)$$

- The portfolio has mean μ and variance

$$\begin{aligned}\mathrm{Var}(R_p) =& \frac{1}{4}(\mathrm{Var}(R_1) + \mathrm{Var}(R_2) + 2\mathrm{Cov}(R_1, R_2)) \\ =& \frac{1}{4}(2\sigma^2 + 2\rho\sigma^2) = \frac{1+\rho}{2}\sigma^2\end{aligned}$$

  - Maximum diversification benefit: $\mathrm{Var}(R_p) = 0$ when $\rho = -1$.
  - No diversification benefit: $\mathrm{Var}(R_p) = \sigma^2$ when $\rho = 1$.

The market model

- A statistical model of stock returns.
- The return on stock i is

$$R_i = R_m + \epsilon_i$$

- R_m is the market return.
- ϵ_i is idiosyncratic risk with mean 0, variance σ^2, and no correlation with R_m.
- For any two stocks $i \neq j$, $\mathrm{Corr}(\epsilon_i, \epsilon_j) = 0$.

Benefits of diversification under the market model

- Construct an equal-weighted portfolio as

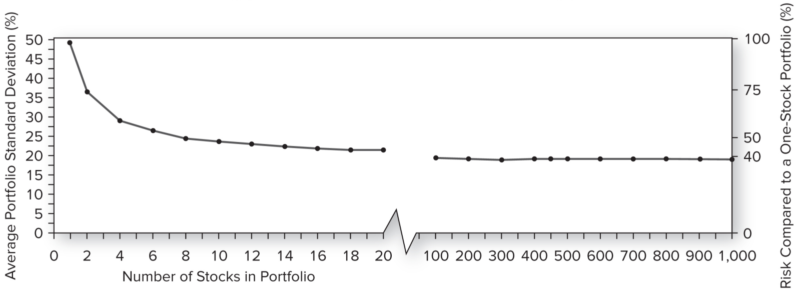
$$\begin{aligned} R_p =& \frac{1}{I}\sum_{i=1}^{I} R_i \\ =& R_m + \frac{1}{I}\sum_{i=1}^{I} \epsilon_i \end{aligned}$$

- The portfolio has variance

$$\begin{aligned} \mathrm{Var}(R_p) =& \mathrm{Var}(R_m) + \mathrm{Var}\left(\frac{1}{I}\sum_{i=1}^{I} \epsilon_i\right) \\ =& \mathrm{Var}(R_m) + \frac{1}{I}\sigma^2 \\ \to& \mathrm{Var}(R_m) \text{ as } I \to \infty \end{aligned}$$

- Idiosyncratic risk disappears through diversification, and only systematic risk (i.e., market return) remains.

How many stocks for a diversified portfolio?



Source: Bodie, Kane, and Marcus (2018, Figure 7.2)

Optimal portfolio diversification

- Define a portfolio return as

$$R_p = \sum_{i=1}^{I} w_i R_i$$

  where $\sum_{i=1}^{I} w_i = 1$.
- Portfolio weight w_i on asset i can be negative (shorting) or greater than 1 (leverage).
- A mean-variance efficient portfolio has
  - Highest expected return among all portfolios with the same variance.
  - Lowest variance among all portfolios with the same expected return.
- Choose portfolio weights to maximize $\mathbb{E}[R_p]$ subject to the constraint that $\mathrm{Var}(R_p) = \sigma^2$.

Two simple portfolio rules

1. Equal-weighted portfolio $w_i = 1/I$.
   Let every man divide his money into three parts, and invest a third in land, a third in business, and a third let him keep by him in reserve. (Talmud, 200 B.C.)
2. Value-weighted portfolio, where w_i is asset i's market cap as a share of total market cap.
   - For example, Vanguard Total Stock Market Index Fund.

- No reason to believe that either strategy is mean-variance efficient.

MSCI World Index

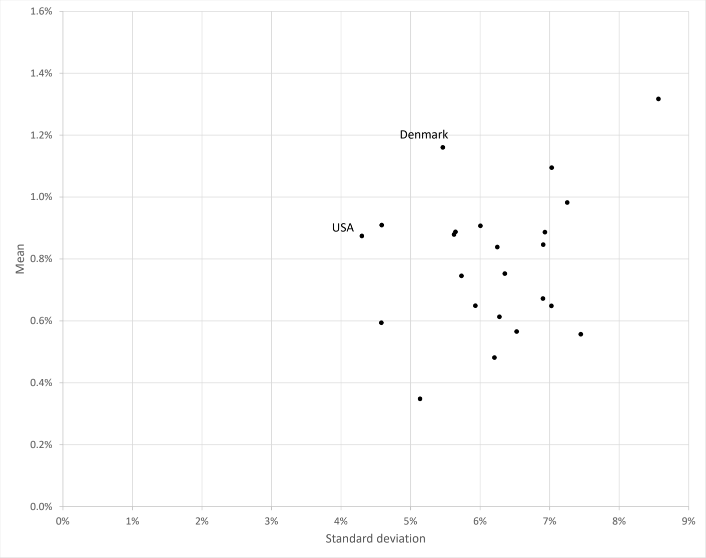
MSCI WORLD INDEX		
DEVELOPED MARKETS		
Americas	Europe & Middle East	Pacific
Canada United States	Austria Belgium Denmark Finland France Germany Ireland Israel Italy Netherlands Norway Portugal Spain Sweden Switzerland United Kingdom	Australia Hong Kong Japan New Zealand Singapore

- Value-weighted portfolio of 23 developed stock markets.
- Can we do better than MSCI by computing the optimal portfolio weights by country?

Example 1: Country-level stock returns (January 1993 to June 2022)

Country	Mean (%)	SD (%)	Sharpe ratio
WORLD	0.74	4.30	0.15
DENMARK	1.16	5.46	0.20
USA	0.87	4.30	0.18
SWITZERLAND	0.91	4.58	0.18
FINLAND	1.32	8.57	0.14
SWEDEN	1.09	7.03	0.14
CANADA	0.89	5.65	0.14
NETHERLANDS	0.88	5.62	0.14
AUSTRALIA	0.91	6.00	0.14
NORWAY	0.98	7.25	0.12
NEW ZEALAND	0.84	6.25	0.12
HONG KONG	0.89	6.93	0.12
FRANCE	0.75	5.73	0.12
UK	0.59	4.58	0.11
SPAIN	0.85	6.91	0.11
GERMANY	0.75	6.36	0.11
BELGIUM	0.65	5.93	0.10
SINGAPORE	0.67	6.90	0.09
PORTUGAL	0.61	6.28	0.08
ITALY	0.65	7.03	0.08
ISRAEL	0.57	6.53	0.07
IRELAND	0.48	6.21	0.06
AUSTRIA	0.56	7.45	0.06
JAPAN	0.35	5.14	0.05

Mean vs. standard deviation across countries



A statistics refresher

- Start with a vector of returns at date t.

$$\mathbf{R}_t = \begin{bmatrix} R_{1,t} \\ \vdots \\ R_{I,t} \end{bmatrix}$$

- Time-series data for $t = 1, \ldots, T$.
- Estimate the mean as

$$\mu = \mathbb{E}[\mathbf{R}_t] = \frac{1}{T} \sum_{t=1}^{T} \mathbf{R}_t$$

Covariance matrix

- Estimate the covariance matrix as

$$\Sigma = \mathbb{E}[(\mathbf{R}_t - \mu)(\mathbf{R}_t - \mu)'] = \frac{1}{T}\sum_{t=1}^{T}(\mathbf{R}_t - \mu)(\mathbf{R}_t - \mu)'$$

  - Row i and column i is $\mathrm{Var}(R_i)$.
  - Row i and column j is $\mathrm{Cov}(R_i, R_j)$.

- In the case of 2 assets,

$$\Sigma = \begin{bmatrix} \mathrm{Var}(R_1) & \mathrm{Cov}(R_1, R_2) \\ \mathrm{Cov}(R_1, R_2) & \mathrm{Var}(R_2) \end{bmatrix}$$

Statistics functions in Excel

- AVERAGE: Function to estimate the mean.
- Estimate the covariance matrix using the covariance function in the Data Analysis ToolPak.
  - First install the Data Analysis ToolPak add-in.
  - http://www.datasciencemadesimple.com/create-covariance-matrix-in-excel/

Matrix functions in Excel

- MINVERSE: Array function to compute the inverse of a covariance matrix.
  - https://www.excelfunctions.net/excel-minverse-function.html
- TRANSPOSE: Array function to compute the transpose of a matrix.
  - https://www.excelfunctions.net/excel-transpose-function.html
- MMULT: Array function to compute the product of two matrices.
  - https://www.excelfunctions.net/excel-mmult-function.html
- To execute an array function, highlight target cells that are the correct dimension and press Ctrl+Shift+Enter.

Example 2: USA and Denmark

- Let's start with a simpler problem of choosing an optimal portfolio of two countries.

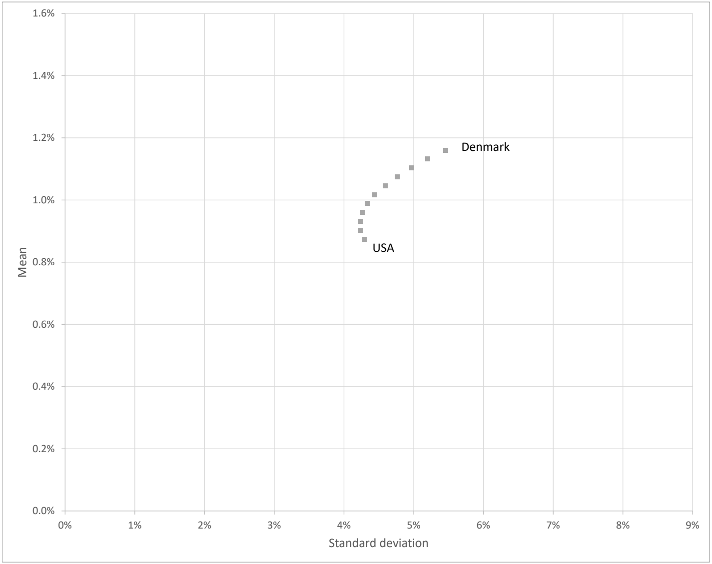
$$R_p = wR_1 + (1 - w)R_2$$

- Using data through June 2022, estimate mean and the covariance matrix.
- Mean and variance of portfolio returns are

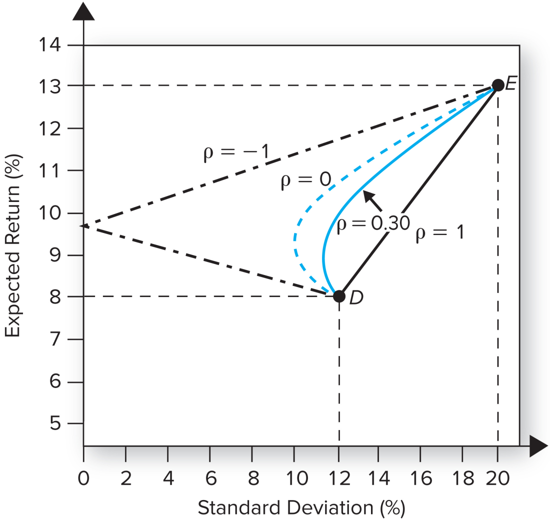
$$\begin{aligned}\mathbb{E}[R_p] =& w\mathbb{E}[R_1] + (1 - w)\mathbb{E}[R_2] \\ \mathrm{Var}(R_p) =& w^2\mathrm{Var}(R_1) + (1 - w)^2\mathrm{Var}(R_2) + 2w(1 - w)\mathrm{Cov}(R_1, R_2)\end{aligned}$$

- Vary w from 0 to 1 to compute all possible combinations.

Efficient frontier for USA and Denmark



Shape of the efficient frontier depends on correlation



Source: Bodie, Kane, and Marcus (2018, Figure 7.5)

Adding the riskless asset

- The analysis so far does not answer:
  - How much to invest in the riskless asset?
  - Which portfolio on the efficient frontier should you choose?
- The riskless rate in June 2022 is 0.08%.
- Write the portfolio return as

$$\begin{aligned} R_p =& (1 - w_1 - w_2) R_b + w_1 R_1 + w_2 R_2 \\ =& R_b + w_1 (R_1 - R_b) + w_2 (R_2 - R_b) \\ =& R_b + \mathbf{w}' (\mathbf{R} - R_b \mathbf{1}) \end{aligned}$$

where

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \mathbf{R} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix}, \mathbf{1} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- The mean is

$$\mathbb{E}[R_p] = R_b + \mathbf{w}'(\mu - R_b\mathbf{1})$$

- The variance is

$$\begin{aligned}
\mathrm{Var}(R_p) =& \mathrm{Var}(R_b + \mathbf{w}'(\mathbf{R} - R_b\mathbf{1})) = \mathrm{Var}(\mathbf{w}'\mathbf{R}) \\
=& \mathbb{E}[(\mathbf{w}'\mathbf{R} - \mathbf{w}'\mu)^2] \\
=& \mathbb{E}[\mathbf{w}'(\mathbf{R} - \mu)(\mathbf{R} - \mu)'\mathbf{w}] \\
=& \mathbf{w}'\mathbb{E}[(\mathbf{R} - \mu)(\mathbf{R} - \mu)']\mathbf{w} = \mathbf{w}'\Sigma\mathbf{w}
\end{aligned}$$

Portfolio-choice problem

- Investor maximizes mean-variance expected utility.

$$\begin{aligned}\mathbb{E}[U] =& \mathbb{E}[R_p] - \frac{\gamma}{2}\mathrm{Var}(R_p) \\ =& R_b + \mathbf{w}'(\mu - R_b\mathbf{1}) - \frac{\gamma}{2}\mathbf{w}'\Sigma\mathbf{w}\end{aligned}$$

- First-order condition:

$$\frac{\partial \mathbb{E}[U]}{\partial \mathbf{w}} = \begin{bmatrix} \frac{\partial \mathbb{E}[U]}{\partial w_1} \\ \vdots \\ \frac{\partial \mathbb{E}[U]}{\partial w_I} \end{bmatrix} = \mu - R_b\mathbf{1} - \gamma\Sigma\mathbf{w} = \mathbf{0}$$

- Solve for the optimal portfolio.

$$\mathbf{w} = \frac{1}{\gamma}\Sigma^{-1}(\mu - R_b\mathbf{1})$$

- The share in the riskless asset is

$$1 - \mathbf{w}'\mathbf{1} = 1 - \sum_{i=1}^{I} w_i$$

Tangency portfolio

- For some level of risk aversion $\widehat{\gamma}$, it is optimal to hold only risky assets and no riskless asset. That is,

$$\widehat{\mathbf{w}} = \frac{1}{\widehat{\gamma}}\Sigma^{-1}(\mu - R_b\mathbf{1})$$

and $1 - \widehat{\mathbf{w}}'\mathbf{1} = 0$.

- Let's solve for $\widehat{\gamma}$.

$$\begin{aligned} 1 =& \widehat{\mathbf{w}}'\mathbf{1} = \frac{1}{\widehat{\gamma}}\mathbf{1}'\Sigma^{-1}(\mu - R_b\mathbf{1}) \\ \widehat{\gamma} =& \mathbf{1}'\Sigma^{-1}(\mu - R_b\mathbf{1}) \end{aligned}$$

- The return on the tangency portfolio is

$$\widehat{R}_p = (1 - \widehat{\mathbf{w}}'\mathbf{1})R_b + \widehat{\mathbf{w}}'\mathbf{R} = \widehat{\mathbf{w}}'\mathbf{R}$$

Portfolio separation theorem

- Every optimal portfolio is a combination of the riskless asset and the tangency portfolio.

$$\mathbf{w} = \frac{1}{\gamma}\Sigma^{-1}(\mu - R_b\mathbf{1}) = \frac{\widehat{\gamma}}{\gamma}\underbrace{\frac{1}{\widehat{\gamma}}\Sigma^{-1}(\mu - R_b\mathbf{1})}_{\widehat{\mathbf{w}}}$$

- Rewrite the return on an optimal portfolio as

$$\begin{aligned} R_p =& (1 - \mathbf{w}'\mathbf{1})R_b + \mathbf{w}'\mathbf{R} \\ =& \left(1 - \frac{\widehat{\gamma}}{\gamma}\widehat{\mathbf{w}}'\mathbf{1}\right) R_b + \frac{\widehat{\gamma}}{\gamma}\widehat{\mathbf{w}}'\mathbf{R} \\ =& \left(1 - \frac{\widehat{\gamma}}{\gamma}\right) R_b + \frac{\widehat{\gamma}}{\gamma}\widehat{R}_p \end{aligned}$$

  - $\gamma > \widehat{\gamma}$: Risk averse investors hold a combination of the riskless asset and the tangency portfolio.
  - $\gamma < \widehat{\gamma}$: Risk tolerant investors short the riskless asset to hold a leveraged position in the tangency portfolio.

Applying the portfolio separation theorem

- Tangency portfolio: 0.48 in USA and 0.52 in Denmark.
- $\widehat{\gamma} = 4.72$.
- If $\gamma = 3$,

$$\mathbf{w} = \frac{\widehat{\gamma}}{\gamma}\widehat{\mathbf{w}} = \frac{4.72}{3}\begin{bmatrix}0.48\\0.52\end{bmatrix} = \begin{bmatrix}0.75\\0.82\end{bmatrix}$$

	γ		
Asset	3	4.72	8
USA	0.75	0.48	0.28
Denmark	0.82	0.52	0.31
Riskless	−0.57	0	0.41

Implications of the portfolio separation theorem

- All investors, regardless of risk aversion, should hold the same portfolio of risky assets.
- More risk averse investors allocate a higher share to the riskless asset.
- Sometimes called the "mutual fund theorem". All investors should hold the same mutual fund of risky assets.
  - Literally interpreted, the mutual fund industry should offer only one type of mutual fund (one size fits all).
- Only portfolios on the capital allocation line are optimal.

Capital allocation line

- Expected return on an optimal portfolio is

$$\mathbb{E}[R_p] = R_b + \frac{\widehat{\gamma}}{\gamma}(\mathbb{E}[\widehat{R}_p] - R_b)$$

- Its standard deviation is

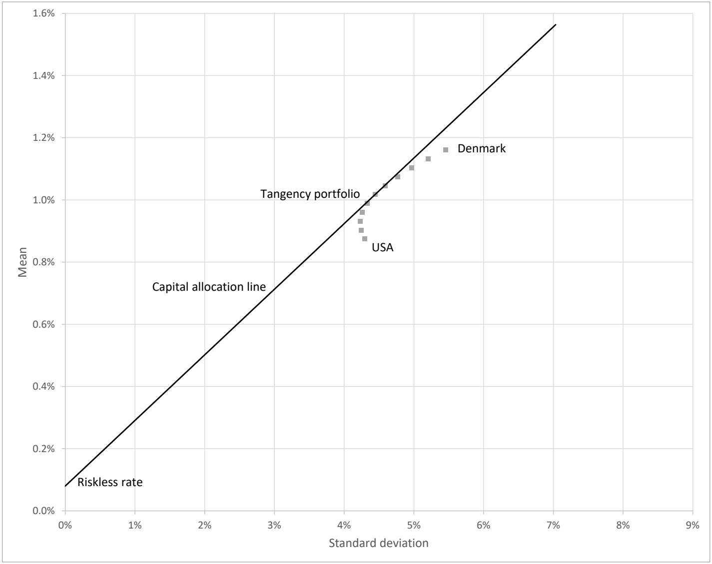
$$\sigma(R_p) = \frac{\widehat{\gamma}}{\gamma}\sigma(\widehat{R}_p)$$

- The capital allocation line is

$$\mathbb{E}[R_p] = R_b + \underbrace{\frac{\mathbb{E}[\widehat{R}_p] - R_b}{\sigma(\widehat{R}_p)}}_{\text{Sharpe ratio}}\,\sigma(R_p)$$

  - Intercept is the riskless rate R_b.
  - Slope is the Sharpe ratio for the tangency portfolio.

Capital allocation line for USA and Denmark



Fidelity Model Portfolios

The primary objective of these Fidelity Model Portfolios is to provide a representation of just one way you might construct a well-diversified portfolio of Fidelity mutual funds based on a particular risk tolerance level.

Asset	Conservative	Moderate	Aggressive
US stocks	14	28	70
Foreign stocks	6	12	30
Bonds	50	45	0
Cash	30	15	0

Source: https://www.fidelity.com/mutual-funds/fidelity-fund-portfolios/overview

Does popular investment advice follow theory?

David Swensen
CIO of Yale

Gretchen Tai
CIO of HP

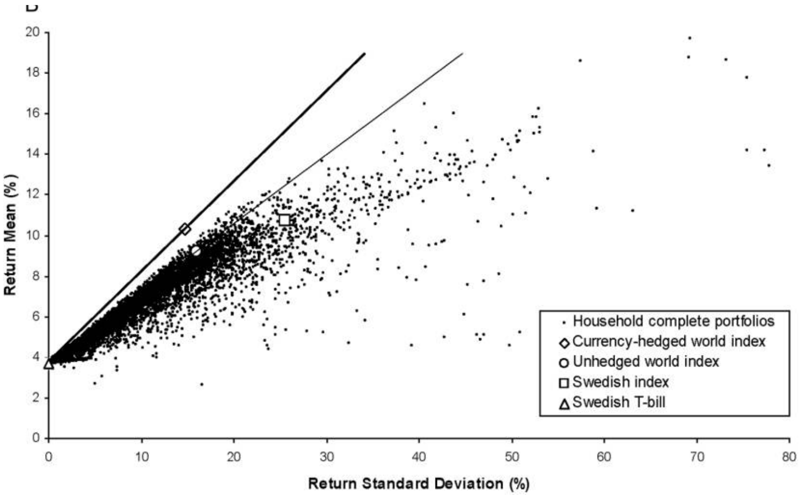
Asset	Swensen	Tai
US stocks	30	29
Foreign developed market stocks	15	26
Emerging market stocks	10	10
Total stocks	55	65
US Treasury bonds	15	4
US TIPS	15	8
Investment-grade bonds	0	10
High-yield bonds	0	8
REITs	15	5
Total bonds	45	35

Source: "3 Investment Gurus Share Their Model Portfolios" (NPR, 10/17/2015)

Why doesn't popular investment advice follow theory?

- Advisors may have different estimates of expected returns and covariance.
  - Swensen and Tai may have different forecasts.
- Our model of optimal portfolio choice may be too simplistic.
  - Investors have more sophisticated preferences than mean-variance.
  - Other sources of risk such as labor income and housing (later in the course).
- Be skeptical of popular investment advice!

How close are households to the capital allocation line?



Source: Calvet, Campbell, and Sodini (2007, *Journal of Political Economy*)

Example 3: Tangency portfolio for all countries

- Let's compute the tangency portfolio for all 23 countries in the MSCI World Index.
- The formula for the tangency portfolio is

$$\widehat{\mathbf{w}} = \frac{1}{\widehat{\gamma}} \Sigma^{-1}(\mu - R_b \mathbf{1})$$
$$\widehat{\gamma} = \mathbf{1}' \Sigma^{-1}(\mu - R_b \mathbf{1})$$

Recipe for computing the tangency portfolio

1. Compute

$$\mathbf{w} = \Sigma^{-1}(\mu - R_b\mathbf{1})$$

2. Compute the sum of the weights

$$\widehat{\gamma} = \mathbf{w}'\mathbf{1} = \sum_{i=1}^{I} w_i$$

3. Divide

$$\widehat{\mathbf{w}} = \frac{1}{\widehat{\gamma}}\mathbf{w}$$

Weights for the tangency portfolio

Country	Mean (%)	SD (%)	Sharpe ratio	Weight
DENMARK	1.16	5.46	0.20	1.02
USA	0.87	4.30	0.18	1.37
SWITZERLAND	0.91	4.58	0.18	0.85
FINLAND	1.32	8.57	0.14	0.20
SWEDEN	1.09	7.03	0.14	−0.02
CANADA	0.89	5.65	0.14	0.01
NETHERLANDS	0.88	5.62	0.14	0.33
AUSTRALIA	0.91	6.00	0.14	0.36
NORWAY	0.98	7.25	0.12	0.14
NEW ZEALAND	0.84	6.25	0.12	0.04
HONG KONG	0.89	6.93	0.12	0.14
FRANCE	0.75	5.73	0.12	0.02
UK	0.59	4.58	0.11	−0.45
SPAIN	0.85	6.91	0.11	0.28
GERMANY	0.75	6.36	0.11	−0.66
BELGIUM	0.65	5.93	0.10	−0.39
SINGAPORE	0.67	6.90	0.09	−0.27
PORTUGAL	0.61	6.28	0.08	−0.29
ITALY	0.65	7.03	0.08	−0.15
ISRAEL	0.57	6.53	0.07	−0.32
IRELAND	0.48	6.21	0.06	−0.52
AUSTRIA	0.56	7.45	0.06	−0.21
JAPAN	0.35	5.14	0.05	−0.47

Lessons

- Suggests a rule-of-thumb to hold assets with higher Sharpe ratios and short assets with lower Sharpe ratios.
- We will discuss types of assets and strategies with high Sharpe ratios.
- Cannot implement extreme short positions in practice.
- Need to constrain weights to be positive to rule out short positions.

Example 4: Tangency portfolio with short sales constraints

- The tangency portfolio has the maximum Sharpe ratio.
- Solve the optimization problem:

$$\max_{\mathbf{w}} \frac{\mathbb{E}[R_p] - R_b}{\sigma(R_p)} = \frac{\mathbf{w}'(\mu - R_b\mathbf{1})}{\sqrt{\mathbf{w}'\Sigma\mathbf{w}}}$$

subject to

$$\begin{aligned} \mathbf{w}'\mathbf{1} &= 1 \\ \mathbf{w} &\geq \mathbf{0} \end{aligned}$$

- Use the Excel solver.

Weights for the tangency portfolio with short sales constraints

Country	Mean (%)	SD (%)	Sharpe ratio	Weight
DENMARK	1.16	5.46	0.20	0.42
USA	0.87	4.30	0.18	0.34
SWITZERLAND	0.91	4.58	0.18	0.22
FINLAND	1.32	8.57	0.14	0.02
SWEDEN	1.09	7.03	0.14	0
CANADA	0.89	5.65	0.14	0
NETHERLANDS	0.88	5.62	0.14	0
AUSTRALIA	0.91	6.00	0.14	0
NORWAY	0.98	7.25	0.12	0
NEW ZEALAND	0.84	6.25	0.12	0
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FRANCE	0.75	5.73	0.12	0
UK	0.59	4.58	0.11	0
SPAIN	0.85	6.91	0.11	0
GERMANY	0.75	6.36	0.11	0
BELGIUM	0.65	5.93	0.10	0
SINGAPORE	0.67	6.90	0.09	0
PORTUGAL	0.61	6.28	0.08	0
ITALY	0.65	7.03	0.08	0
ISRAEL	0.57	6.53	0.07	0
IRELAND	0.48	6.21	0.06	0
AUSTRIA	0.56	7.45	0.06	0
JAPAN	0.35	5.14	0.05	0

Lessons

- Still hold assets with higher Sharpe ratios.
- However, the tangency portfolio is sparse (i.e., lots of zeros).
- Optimal diversification (with short sales constraints) can result in concentrated portfolios.

Efficient frontier with short sales constraints

- A mean-variance efficient portfolio has the highest expected return among all portfolios with the same variance.
- Solve the optimization problem:

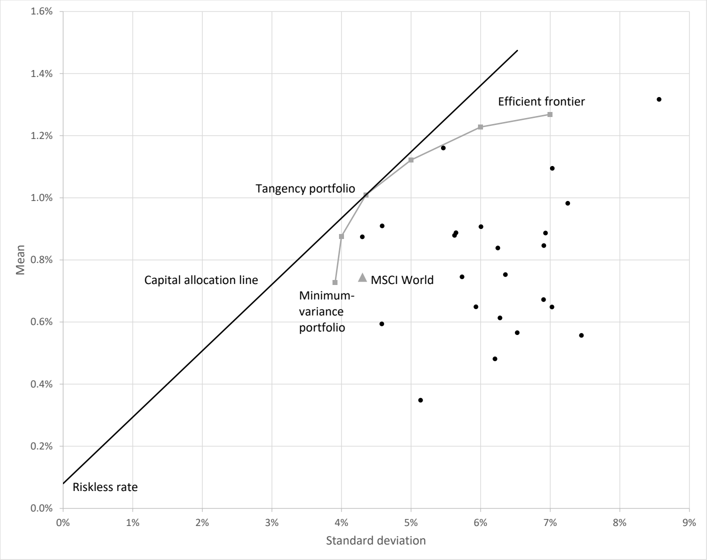
$$\max_{\mathbf{w}} \ \mathbb{E}[R_p] = R_b + \mathbf{w}'(\mu - R_b\mathbf{1})$$

subject to

$$\begin{aligned} \sigma(R_p) =& \sqrt{\mathbf{w}'\Sigma\mathbf{w}} = a \\ \mathbf{w}'\mathbf{1} =& 1 \\ \mathbf{w} \geq& \mathbf{0} \end{aligned}$$

- Repeat for different values of a.

Efficient frontier with short sales constraints



Minimum-variance portfolio with short sales constraints

- The minimum-variance portfolio has the lowest variance among all portfolios of risky assets.
- Solve the optimization problem:

$$\min_{\mathbf{w}} \ \sigma(R_p) = \sqrt{\mathbf{w}'\Sigma\mathbf{w}}$$

subject to

$$\mathbf{w}'\mathbf{1} = 1$$
$$\mathbf{w} \geq \mathbf{0}$$

Vocabulary test

- A portfolio is mean-variance efficient if it has the highest expected return among portfolios with the same variance.
- The efficient frontier is a curve that connects all portfolios that are mean-variance efficient.
- The minimum-variance portfolio is a portfolio on the efficient frontier that has the lowest variance.
- The tangency portfolio is a portfolio on the efficient frontier that has the highest Sharpe ratio.
- The capital allocation line connects the tangency portfolio and the riskless rate.

Summary

- Hold a diversified portfolio. You should not bear idiosyncratic risk.
- The market portfolio is not necessarily optimal.
- The optimal portfolio has the highest Sharpe ratio. Look for strategies with high Sharpe ratios.

Next steps

- Now that we have a theory of demand, combine with supply for an asset pricing model.
- Advanced topics addressed in ECO 462 (Portfolio Theory and Asset Management).
  - Covariance matrix hard to estimate for large-scale problems.
  - Portfolio constraints such as limits on shorting and leverage.
  - Adjusting portfolio weights to take advantage of time-varying mean and variance.